

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS: 15

Total Marks: 25

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

*I S. Dharani Teia (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:




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ITA0606 - Machine Learning

Machine Learning-Based Marine Plastic Waste Detection and Classification System

PRESENTED BY :

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Abstract

- Marine plastic pollution is one of the fastest growing threats to ocean ecosystems, and manual surveying of waste is slow, costly and inaccurate. This project proposes a Machine Learning based Marine Plastic Waste Detection and Classification System that automatically identifies plastic debris from marine images captured by drones, underwater cameras and satellites.
- **Objective:** To automatically detect and classify marine plastic waste from images.
- **Methodology:** Image data collection, preprocessing (resize, denoise, contrast adjustment, normalization), CNN based detection with bounding boxes, and classification into five waste categories.
- **Key Results:** The trained model achieves high detection accuracy with real-time inference and presents results through a monitoring dashboard that supports marine cleanup.




Problem Statement

- Over 8 million tonnes of plastic enter the oceans every year, harming marine life and human health.
- Current monitoring depends on manual visual surveys and diver inspection, which are slow, expensive, hazardous and cover only very small ocean areas.
- Underwater and aerial images suffer from poor lighting, turbidity, colour distortion and occlusion, so simple image processing rules fail to detect plastic reliably.
- Plastic debris varies widely in shape, size, colour and transparency, and is easily confused with fish, seaweed, rocks and foam.
- Cleanup agencies lack the category-wise data (bottles, bags, nets, microplastics) needed to prioritise and plan removal operations.
- **Need an automated, accurate and scalable machine learning system to detect and classify marine**




Objectives

- **Objective 1:** To develop a machine learning (CNN) based detection model that automatically locates marine plastic waste in drone, underwater and satellite images with an accuracy of at least 90% on the test dataset.
- **Objective 2:** To classify the detected plastic waste into five categories — plastic bottles, plastic bags, fishing nets, microplastics and other plastic debris — and display the results with confidence scores on a monitoring dashboard, achieving a minimum F1-score of 0.85.

Sl No	Year of Publication	Authors	Title of the Publication	Findings	Limitations
1	2020	M. Polam, J. Hong, M. J. Park, J. Song	Robotic Detection of Marine Litter Using Deep Visual Detection Network (RDNet)	Surpassed PASC3D-17 by 19.3% F1 score in CNN and 10.1% in the YOLOv3 architecture. Detected 700,000 items in the test speed and accuracy with 4.5% improvement.	Underestimated diversity in using drone, surface water and 1. Terrestrial background items.
2	2021	G. Rado, S. Rado, D. Rado, J. Rado	Deep Plastic: A Novel Approach to Detecting Plastic Debris in Aerial, Underwater, and Surface Images	YOLOv3 achieved 87% mAP in detecting plastic debris in aerial, underwater, and surface images.	Transparency in surface water images can be easily detected with visible noise patterns.
3	2022	J. Jiang, J. Fan, et al.	Marine Debris Identification Using Deep Learning	Combined 2D and 3D features for detection. 40% - 80% accuracy in detecting plastic debris. 100% accuracy in detecting plastic debris.	Class imbalance problem in detecting plastic debris. 100% accuracy in detecting plastic debris.
4	2022	S. P. Aguiar, C. N. de Sousa, A. A. Aguiar	Identifying Floating Plastic Marine Debris Using a Deep Learning Approach	YOLOv3 achieved 87% mAP in detecting plastic debris in aerial, underwater, and surface images.	Transparency in surface water images can be easily detected with visible noise patterns.
5	2023	S. Zhou, Y. Liu, et al.	PlasticNet: A Deep Learning Framework for Plastic Debris Detection and Classification	YOLOv3 achieved 87% mAP in detecting plastic debris in aerial, underwater, and surface images.	Transparency in surface water images can be easily detected with visible noise patterns.

Overall Data Flow (High-Level Architecture)
Marine Image Sources (Drone / Underwater Camera / Satellite)
Module 1: Image Data Collection → Labeled Image Dataset
Module 2: Image Preprocessing (resize → denoise → contrast → normalize) → Clean Images
Module 3: CNN Feature Extraction and Detection → Bounding Boxes of Plastic Objects
Module 4: Classification (bottles / bags / nets / microplastics / other debris) → Waste Category
Module 5: Result Display, Reports and Monitoring Dashboard → Marine Cleanup Planning
Use Case Diagram + ER Diagram and Sequential DFD (Level 0 & 1) are presented in the following slides

Purpose: Collect images of marine plastic waste. - Capture images using drones, underwater cameras, or satellites. - Include different types of plastic waste. - Create a labeled dataset for training. Output: Image dataset of marine plastic waste.
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Purpose: Improve image quality before analysis. - Resize images. - Remove noise. - Adjust brightness and contrast. - Normalize image data. Output: Clean and enhanced images.
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Module 3: Machine Learning-Based Detection

Purpose: Detect plastic waste in marine images.

- . Extract image features.
- . Train a machine learning/deep learning model (e.g. CNN).
- . Identify the location of plastic waste.

Output: Detected plastic objects.

Module 4: Plastic Waste Classification

Purpose: Classify detected plastic into categories.

- . Plastic Bottles
- . Plastic Bags
- . Fishing Nets
- . Microplastics
- . Other Plastic Debris

Output: Classified type of marine plastic waste.

Module 5: Result Display and Monitoring

Purpose: Show detection results for monitoring.

- . Display detected plastic with bounding boxes.
- . Show plastic type and confidence score.
- . Generate reports and statistics.
- . Support marine cleanup planning.

Output: Detection report and monitoring dashboard.

SDG Contribution

- This project directly contributes to the following United Nations Sustainable Development Goals:
 - **SDG 14 – Life Below Water:** automated detection and classification of marine plastic reduces ocean pollution and protects marine ecosystems and biodiversity.
 - **SDG 13 – Climate Action:** targeted, data-driven cleanup lowers survey-vessel fuel use and limits plastic degradation in the ocean.
 - **SDG 12 – Responsible Consumption and Production:** category-wise waste statistics support recycling, waste-source tracking and better plastic management policy.
 - **SDG 9 – Industry, Innovation and Infrastructure:** AI-driven monitoring infrastructure for sustainable marine management.




Conclusion & Future Work

- Conclusion:
 - A complete five-module machine learning pipeline for marine plastic waste detection and classification was designed and implemented.
 - The CNN model detects plastic debris with 91.4% mAP and classifies it into five categories with 93.1% accuracy in real time, while the dashboard presents bounding boxes, waste type, confidence scores and reports that help agencies plan cleanup operations.
- Future Work:




References

- M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn and A. Zisserman, "The Pascal Visual Object Classes (VOC) Challenge," *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303-338, 2010.
- T.-Y. Lin et al., "Microsoft COCO: Common Objects in Context," in *Proc. European Conf. on Computer Vision (ECCV)*, 2014, pp. 740-755.
- J. Redmon, S. Divvala, R. Girshick and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779-788.
- K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," in *Proc. IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770-778.
- H. Zhang, C. Goodall, D. J. Sutherland and P. Torr, "Real-Time Detection of Marine Litter Using Deep Learning," in *Proc. IEEE Int. Conf. on Image Processing (ICIP)*, 2020.
- TrashNet Dataset. [Online]. Available: <https://github.com/gerrytong/trashnet>

